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Studierichting: Informatica
Oefeningenreeks 2B nr. 5.3

Oefening: $\mathrm{Bgtan}(x) + \mathrm{Bgtan}\left(\frac{1}{x}\right)$

Stel $x = \tan(\alpha)$ met $\alpha \in \left]0, \frac{\pi}{2}\right[$

Dan geldt $\frac{1}{x} = \frac{1}{\tan(\alpha)} = \cot(\alpha) = \tan\left(\frac{\pi}{2} - \alpha\right)$

Dus $\mathrm{Bgtan}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \alpha$

$\Rightarrow \mathrm{Bgtan}(x) + \mathrm{Bgtan}\left(\frac{1}{x}\right) = \alpha + \left(\frac{\pi}{2} - \alpha\right) = \frac{\pi}{2}$

Conclusie:

$$\forall x > 0 : \mathrm{Bgtan}(x) + \mathrm{Bgtan}\left(\frac{1}{x}\right) = \frac{\pi}{2}$$

References